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Structural Loads Analysis: Acceleration, Pressure, Acoustic, and Spin Loads

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1. Load Case Summary

The AMSR-UBSCLS spear experiences 5 distinct load cases:

Load Case	Phase	Duration	Max Load
LC1: Smart Rope Pull	Surface contact	0.04 s	50,040 N per rope
LC2: Hydrodynamic drag	Underwater ascent	1-4 s	4,063 N
LC3: Transonic buffet	Mach 0.8-1.2	0.5-2 s	Variable, ± 1500 N
LC4: Second stage firing	Post-apogee	28 s	2754 N axial
LC5: Centrifugal (spin)	Entire flight	continuous	Blades: 1.44 MPa

Critical load case: LC1 — Smart Rope Pull at surface

1.1 LC1 — Smart Rope Pull Analysis

For $n=6$ vessels, $F_{\text{rope_max}} = 50,040$ N each, all attached to spear base (aft 50mm). Total axial force applied to spear: $F_{\text{total}} = 6 * 50040 = 300,240$ N

Average axial stress at spear cross-section:

$$\begin{aligned}\sigma_{\text{axial}} &= F_{\text{total}} / A_{\text{cross}} = 300240 / (\pi * (0.04)^2 - \pi * (0.036)^2) \\ &= 300240 / (5.03\text{e-}3 - 4.07\text{e-}3) \\ &= 300240 / 9.6\text{e-}4 = 312.75 \text{ MPa}\end{aligned}$$

ZrB2-SiC tensile strength: 850 MPa **Safety factor: $850/313 = 2.72$** — adequate for a single-use system.

For hollow shell (outer $R=40\text{mm}$, inner $R=36\text{mm}$, wall $t=4\text{mm}$, $L=640\text{mm}$):

Peak bending stress at attachment point (if ropes not perfectly centered): Assume 1mm misalignment: $e = 0.001$ m Bending moment: $M = F_{\text{total}} * e = 300240 * 0.001 = 300$ N · m Bending stress: $\sigma_{\text{bend}} = M * R / I = 300 * 0.040 / (\pi/64 * (0.08^4 - 0.072^4)) = 54.8$ MPa

Combined: $\sigma_{\text{total}} = 313 + 55 = 368$ MPa < 850 MPa → **Safety factor 2.31.**

1.2 LC2 — Hydrodynamic Drag Loading

Max drag force (underwater at 22 m/s): 4063 N (Vol 03) This is a compressive axial load on the spear. ZrB2-SiC compressive strength: 1800 MPa » any achieved stress.

1.3 LC3 — Transonic Buffet

Oscillatory loading at Mach 0.8-1.2 from shock-boundary layer interaction. Frequency: typically 50-200 Hz (structural resonance risk)

Natural frequency of spear:

$$\begin{aligned}f_n &= 0.561 / L^2 * \sqrt{E * I / (\rho * A)} = 0.561 / 0.64^2 * \sqrt{250\text{e}9 * 1.1\text{e-}6 / (6090 * 9.6\text{e-}4)} \\ &= 1.369 * \sqrt{2.75\text{e}5 / 5.85} = 1.369 * \sqrt{47009} = 297 \text{ Hz}\end{aligned}$$

Buffet frequency range (50-200 Hz) does not overlap with structural natural frequency (297 Hz): **No resonance risk. KNbO3-BaTiO3 dampers provide additional margin.**

2. Fatigue Analysis

For a ceramic material (ZrB₂-SiC), cyclic fatigue does not apply in the same way as metals. Ceramics fail by: 1. **Quasi-static fracture** when stress exceeds fracture toughness locally 2. **Slow crack growth** under sustained load (stress corrosion in water)

Stress corrosion in seawater: ZrB₂-SiC: no water-reaction mechanism. Stress corrosion not applicable (inorganic, non-oxide). **No fatigue limit concern.**

3. FEA Summary (Simplified Beam Model)

For Scale 3 (Small, 10 kg) spear with 6-rope pull:

Location	Axial Stress (MPa)	Shear Stress (MPa)	Safety Factor
Rope attachment (aft)	$313 + 55 = 368$	8	2.31
Mid-span	85	22	10.0
Nose cone join	12	15	51
Screw blade root	1.44	0.9	590

All regions have safety factor > 2.0. The rope attachment zone is limiting.

V2.0 design modification: Thicken rope attachment flange wall from 4mm to 7mm. New safety factor at attachment: $850 / (312.75 * 4/7) = 4.05$ — improved.

Citations (Vol 21)

- Timoshenko, S., Strength of Materials, Van Nostrand 1958 (beam theory)
- Lawn, B.R., Fracture of Brittle Solids, 2nd Ed, Cambridge UP 1993
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- NASA SP-8072, “Acoustic Loads Generated by the Propulsion System,” 1971
- Antunes, J., Axisa, F., Villard, B., Quinio, D., “A Theoretical Analysis of Dynamic Contact Forces in Hydraulic Joints,” ASME 1992

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